

Notes & Steps



Key idea

The derivative $f'(x)$ is found using differentiation by first principles:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This gives the gradient function of $f(x)$.

Quick checks

- ▶ For $f(x) = x^2$: $f'(x) = 2x$
- ▶ For $f(x) = 4x$: $f'(x) = 4$
- ▶ For $f(x) = x^2 + 5$: $f'(x) = 2x$
- ▶ Constants differentiate to zero

Steps

1. Write $f(x+h)$ by replacing x with $x+h$.
2. Expand and simplify $f(x+h) - f(x)$.
3. Divide by h : $\frac{f(x+h) - f(x)}{h}$.
4. Take the limit as $h \rightarrow 0$ (the terms with h disappear).

Common mistake

Forgetting to expand $(x+h)^2$ correctly. Remember $(x+h)^2 = x^2 + 2xh + h^2$, not $x^2 + h^2$. Also common: forgetting that the constant term cancels out in step 2.

Notes & Steps



Example 1: $f(x) = x^2$

$$\begin{aligned}f(x+h) &= (x+h)^2 = x^2 + 2xh + h^2 \\f(x+h) - f(x) &= (x^2 + 2xh + h^2) - x^2 = 2xh + h^2 \\ \frac{f(x+h) - f(x)}{h} &= \frac{2xh + h^2}{h} = 2x + h \\ \text{As } h \rightarrow 0: f'(x) &= 2x\end{aligned}$$

Example 2: $f(x) = 3x^2 + 2x$

$$\begin{aligned}f(x+h) &= 3(x+h)^2 + 2(x+h) = 3x^2 + 6xh + 3h^2 + 2x + 2h \\f(x+h) - f(x) &= 6xh + 3h^2 + 2h = h(6x + 3h + 2) \\ \frac{f(x+h) - f(x)}{h} &= 6x + 3h + 2 \\ \text{As } h \rightarrow 0: f'(x) &= 6x + 2\end{aligned}$$

Example 3: $f(x) = x^3$

$$\begin{aligned}f(x+h) &= (x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3 \\f(x+h) - f(x) &= 3x^2h + 3xh^2 + h^3 = h(3x^2 + 3xh + h^2) \\ \frac{f(x+h) - f(x)}{h} &= 3x^2 + 3xh + h^2 \\ \text{As } h \rightarrow 0: f'(x) &= 3x^2\end{aligned}$$

Try these

1. $f(x) = 2x^2$ — find $f'(x)$ by first principles
2. $f(x) = x^2 + 4x$ — find $f'(x)$ by first principles
3. $f(x) = 5x^2 - 3x + 2$ — find $f'(x)$ by first principles